

Competency

- To simplify the expressions involving indices and solve the exponential equations

Learning Outcomes

- To simplify the expressions by using the laws of indices related to negative and fractional power
- To solve the exponential equations

Level-wise learning objectives

S.N.	LEVELS	OBJECTIVES
1.	Knowledge (K)	- To identify coefficient, base and power of expression - To recall the laws of indices - To express the product/quotient of expressions having same base in terms of single base
2.	Understanding (U)	- To evaluate the numerical problems by using laws of indices - To simplify/prove the simple given expressions - To solve the exponential equations
3.	Application (A)	- To simplify/prove the given rational expressions (involving roots as well) by applying the laws of indices - To solve the exponential equations of the quadratic form
4.	High Ability (HA)	- To prove the rational expression under the given condition/s. - To prepare the report about the use of indices

Required Teaching Materials/ Resources

Chart papers with laws of indices, scissors, ruler, glue-stick and computer/projector if possible

Pre-knowledge: Laws of indices, basic operations

A. Indices**Teaching Activities**

- Give the practical examples of use of laws of indices.
For example
 - The cost of 1 kg of apple is Rs 125. Find the cost of 5 kg of apples by using the product law of indices. For, $5 \times 125 = 5^1 \times 5^3 = 5^{1+3} = 5^4 = 625$
 - Divide 64 copies are equally among 4 friends by using the quotient law of indices. For, $\frac{64}{4} = \frac{2^6}{2^2} = 2^{6-2} = 2^4 = 16$
- Recall of indices by presenting in chart paper with proper examples
- Table of law of indices discuss, give the way of solving the various problems and involve the students in solving the problems from exercise
- Under given condition, prove the expressions and give the same type problems to the students and tell them to prove in the class.

Solution:

$$\begin{aligned}
 \text{a)} \quad & \frac{p + (pq^2)^{\frac{1}{3}} + (p^2q)^{\frac{1}{3}}}{p - q} \times \left(1 - \frac{q^{\frac{1}{3}}}{p^{\frac{1}{3}}}\right) \\
 &= \frac{p + p^{\frac{1}{3}}q^{\frac{2}{3}} + p^{\frac{2}{3}}q^{\frac{1}{3}}}{p - q} \times \left(\frac{p^{\frac{1}{3}} - q^{\frac{1}{3}}}{p^{\frac{1}{3}}}\right) \\
 &= \frac{\cancel{p^{\frac{1}{3}}}(p^{\frac{2}{3}} + q^{\frac{2}{3}} + p^{\frac{1}{3}}q^{\frac{1}{3}})}{p - q} \times \left(\frac{p^{\frac{1}{3}} - q^{\frac{1}{3}}}{\cancel{p^{\frac{1}{3}}}}\right) \\
 &= \frac{(p^{\frac{1}{3}} - q^{\frac{1}{3}})[(p^{\frac{1}{3}})^2 + p^{\frac{1}{3}}q^{\frac{1}{3}} + (q^{\frac{1}{3}})^2]}{p - q} = \frac{(p^{\frac{1}{3}})^3 - (q^{\frac{1}{3}})^3}{p - q} = \frac{p - q}{p - q} = 1
 \end{aligned}$$

$$\begin{aligned}
 \text{b)} \quad & \left(x^{\frac{b}{b-c}}\right)^{\frac{1}{b-a}} \times \left(x^{\frac{c}{c-a}}\right)^{\frac{1}{c-b}} \times \left(x^{\frac{a}{a-b}}\right)^{\frac{1}{a-c}} \\
 &= x^{\frac{b}{(b-c)(b-a)}} \times x^{\frac{c}{(c-a)(c-b)}} \times x^{\frac{a}{(a-b)(a-c)}} \\
 &= x^{\frac{b}{(b-c)(b-a)} + \frac{c}{(c-a)(c-b)} + \frac{a}{(a-b)(a-c)}} \\
 &= x^{\frac{-b}{(b-c)(a-b)} - \frac{c}{(c-a)(b-c)} - \frac{a}{(a-b)(c-a)}} = x^{\frac{-b(c-a) - c(a-b) - a(b-c)}{(a-b)(b-c)(c-a)}} = x^{\frac{-bc + ab - ca + bc - ab + ca}{(a-b)(b-c)(c-a)}} = x^0 = 1
 \end{aligned}$$

$$\begin{aligned}
 \text{c)} \quad & \sqrt[x+y]{\frac{a^{x^2}}{a^{y^2}}} \times \sqrt[y+z]{\frac{a^{y^2}}{a^{z^2}}} \times \sqrt[z+x]{\frac{a^{z^2}}{a^{x^2}}} \\
 &= \sqrt[x+y]{a^{\frac{x^2 - y^2}{y^2}}} \times \sqrt[y+z]{a^{\frac{y^2 - z^2}{z^2}}} \times \sqrt[z+x]{a^{\frac{z^2 - x^2}{x^2}}} \\
 &= a^{\frac{x^2 - y^2}{x+y}} \times a^{\frac{y^2 - z^2}{y+z}} \times a^{\frac{z^2 - x^2}{z+x}} = a^{x - y + y - z + z - x} = a^0 = 1
 \end{aligned}$$

4. (a) If $xyz = 1$, prove that $\frac{1}{1+x+y^{-1}} + \frac{1}{1+y+z^{-1}} + \frac{1}{1+z+x^{-1}} = 1$.

(b) If $a + b + c = 0$, prove that $\frac{1}{1+x^a+x^{-b}} + \frac{1}{1+x^b+x^{-c}} + \frac{1}{1+x^c+x^{-a}} = 1$

Solution:

(a) Here, $xyz = 1$

$$\begin{aligned}
 \text{L.H.S} &= \frac{1}{1+x+y^{-1}} + \frac{1}{1+y+z^{-1}} + \frac{1}{1+z+x^{-1}} \\
 &= \frac{1}{1+x+\frac{1}{y}} + \frac{1}{1+y+\frac{1}{z}} + \frac{1}{1+z+\frac{1}{x}} \\
 &= \frac{1}{\frac{y+xy+1}{y}} + \frac{1}{\frac{z+yz+1}{z}} + \frac{1}{\frac{x+zx+1}{x}} \\
 &= \frac{z \times y}{z(y+xy+1)} + \frac{z}{z+yz+1} + \frac{x}{x+zx+1} \\
 &= \frac{yz}{yz+xyz+z} + \frac{z}{z+yz+1} + \frac{x}{x+zx+1} \\
 &= \frac{yz}{yz+1+z} + \frac{z}{z+yz+1} + \frac{x}{x+zx+1} \\
 &= \frac{x(yz+z)}{x(yz+z+1)} + \frac{x}{x+zx+1} = \frac{xyz+zx}{xyz+zx+x} + \frac{x}{x+zx+1} = \frac{1+zx+x}{1+zx+x} = 1 = \text{RHS}
 \end{aligned}$$

(b) Here, $a + b + c = 0$

$$\begin{aligned}
 \text{LHS} &= \frac{1}{1 + x^a + x^{-b}} + \frac{1}{1 + x^b + x^{-c}} + \frac{1}{1 + x^c + x^{-a}} = 1 \\
 &= \frac{1}{1 + x^a + \frac{1}{x^b}} + \frac{1}{1 + x^b + \frac{1}{x^c}} + \frac{1}{1 + x^c + \frac{1}{x^a}} \\
 &= \frac{x^b \times x^c}{x^c(x^b + x^{a+b} + 1)} + \frac{x^c}{x^c + x^{b+c} + 1} + \frac{x^a}{x^a + x^{c+a} + 1} \\
 &= \frac{x^{b+c}}{x^{b+c} + x^{a+b+c} + x^c} + \frac{x^c}{x^{b+c} + x^c + 1} + \frac{x^a}{x^{c+a} + x^a + 1} \\
 &= \frac{x^{b+c}}{x^{b+c} + x^0 + x^c} + \frac{x^c}{x^{b+c} + x^c + 1} + \frac{x^a}{x^{c+a} + x^a + 1} \\
 &= \frac{x^{b+c} + x^c}{x^{b+c} + x^c + 1} + \frac{x^a}{x^{c+a} + x^a + 1} \\
 &= \frac{x^a(x^{b+c} + x^c)}{x^a(x^{b+c} + x^c + 1)} + \frac{x^a}{x^{c+a} + x^a + 1} \\
 &= \frac{x^{a+b+c} + x^{c+a}}{x^{a+b+c} + x^{c+a} + x^a} + \frac{x^a}{x^{c+a} + x^a + 1} \\
 &= \frac{x^0 + x^{c+a}}{x^0 + x^{c+a} + x^a} + \frac{x^a}{x^{c+a} + x^a + 1} \\
 &= \frac{1 + x^{c+a}}{1 + x^{c+a} + x^a} + \frac{x^a}{x^{c+a} + x^a + 1} = \frac{x^{c+a} + x^a + 1}{x^{c+a} + x^a + 1} = 1 = \text{R.H.S}
 \end{aligned}$$

5. a) If $x = 2^{\frac{1}{3}} + 2^{-\frac{1}{3}}$, prove that: $2x^3 - 6x = 5$.

b) If $x - 2 = 3^{\frac{1}{3}} + 3^{\frac{2}{3}}$, show that: $x(x^2 - 6x + 3) = 2$.

Solution:

a) Here, $\frac{1}{3} + 2^{-\frac{1}{3}}$

Cubing on both sides, we get,

$$x^3 = \left(2^{\frac{1}{3}} + 2^{-\frac{1}{3}}\right)^3$$

$$\text{or, } x^3 = \left(2^{\frac{1}{3}}\right)^3 + \left(2^{-\frac{1}{3}}\right)^3 + 3 \times 2^{\frac{1}{3}} \times 2^{-\frac{1}{3}} \left(2^{\frac{1}{3}} + 2^{-\frac{1}{3}}\right) \quad [\because (a+b)^3 = a^3 + b^3 + 3ab(a+b)]$$

$$\text{or, } x^3 = 2 + 2^{-1} + 3 \times 1 \times x \quad [\because 2^{\frac{1}{3}} + 2^{-\frac{1}{3}} = x]$$

$$\text{or, } x^3 = 2 + \frac{1}{2} + 3x$$

$$\text{or, } x^3 = \frac{4 + 1 + 6x}{2}$$

$$\text{or, } 2x^3 - 6x = 5 \text{ proved}$$

b) Here,

$$x - 2 = 3^{\frac{1}{3}} + 3^{\frac{2}{3}}$$

Cubing on both sides, we get

$$(x-2)^3 = \left(3^{\frac{1}{3}} + 3^{\frac{2}{3}}\right)^3$$

or, $x^3 - 2^3 - 3 \times x \times 2(x-2) = \left(3^{\frac{1}{3}}\right)^3 + \left(3^{\frac{2}{3}}\right)^3 + 3 \times 3^{\frac{1}{3}} \times 3^{\frac{2}{3}} \left(3^{\frac{1}{3}} + 3^{\frac{2}{3}}\right)$

or, $x^3 - 8 - 6x(x-2) = 3 + 3^2 + 3 \times 3^1 \times (x-2)$

or, $x^3 - 8 - 6x^2 + 12x = 12 + 9x - 18$

or, $x^3 - 6x^2 + 3x = 2$

$\therefore x(x^2 - 6x + 3) = 2$ proved.

6. Solve: $2^{x+3} \times 3^{x+2} = 432$

Solution:

Here, $2^{x+3} \times 3^{x+2} = 432$

or, $2^x \times 2^3 \times 3^x \times 3^2 = 432$

or, $(2 \times 3)^x \times 8 \times 9 = 432$

or, $6^x = 6 \quad \therefore x = 1$

7. a) If $x^a = y$, $y^b = z$ and $z^c = x$, prove that $abc = 1$

b) If $a^{\frac{1}{x}} = b^{\frac{1}{3}}$ and $ab = 1$, prove that $x + 3 = 0$.

c) If $a^x = b^y$ and $a^y = b^x$, show that $x = y$

Solution:

a) Here, $x^a = y$, $y^b = z$ and $z^c = x$.

Now, $x^a = y$ or, $(z^c)^a = y$ [$\therefore x = z^c$]

or, $z^{ca} = y$ or, $(y^b)^{ca} = y$ [$\therefore z = y^b$]

or, $y^{abc} = y^1 \quad \therefore abc = 1$ proved.

b) Here, $a^{\frac{1}{x}} = b^{\frac{1}{3}}$ or, $\left(a^{\frac{1}{x}}\right)^x = \left(b^{\frac{1}{3}}\right)^x$

$\therefore a = b^{\frac{x}{3}}$

Now, $ab = 1$ or, $b^{\frac{x}{3}} \cdot b = 1$

or, $b^{\frac{x}{3}+1} = b^0$ or, $x + 3 = 0$ proved.

c) Here, $a^x = b^y$ (i)
 $a^y = b^x$ or, $b^x = a^y$ (ii)

Multiplying equⁿ (i) and (ii), we get

$a^x \cdot b^x = b^y \cdot a^y$

or, $(ab)^x = (ab)^y$

$\therefore x = y$ proved.

8. a) If $(a^{-1} + b^{-1})(a + b)^{-1} = a^m b^n$, prove that $a^{m-n} = 1$

b) If $\left(\frac{m^{-1}n^2}{m^2n^4}\right)^7 \div \left(\frac{m^3n^{-5}}{m^{-2}n^3}\right)^{-5} = m^x n^y$, prove that $m^{x-2y} = 1$.

Solution:

a) Here, $(a^{-1} + b^{-1})(a + b)^{-1} = a^m b^n$

or, $\left(\frac{1}{a} + \frac{1}{b}\right)\left(\frac{1}{a+b}\right) = a^m b^n$

or, $\left(\frac{b+a}{ab}\right)\left(\frac{1}{a+b}\right) = a^m b^n$

or, $a^{-1}b^{-1} = a^m b^n \quad \therefore m = -1, n = -1$

Again, $a^{m-n} = a^{-1-(-)} = a^0 = 1$ proved

$$b) \left(\frac{m^{-1}n^2}{m^2n^4} \right)^7 \div \left(\frac{m^3n^{-5}}{m^{-2}n^3} \right)^{-5} = m^x n^y$$

$$\text{or, } (m^{-3}n^6)^7 \div (m^5n^{-8})^{-5} = m^x n^y$$

$$\text{or, } m^{-21}n^{42} \div m^{-25}n^{40} = m^x n^y$$

$$\text{or, } m^4n^2 = m^x n^y$$

$$\therefore x = 4 \text{ and } y = 2$$

$$\text{Again, } m^{x-2y} = m^{4-2 \times 2} = m^0 = 1 \text{ proved}$$

9. If $x^m \cdot x^n = (x^m)^n$, prove that $x^{m(m-2)} \times x^{n(m-2)} = 1$

Solution:

$$\text{Here, } x^m \cdot x^n = (x^m)^n$$

$$\text{or, } x^{m+n} = x^{mn}$$

$$\text{or, } m + n = mn \dots\dots (i)$$

Now,

$$\text{LHS} = x^{m(n-2)} \times x^{n(m-2)}$$

$$= x^{mn - 2m + mn - 2n}$$

$$= x^{2mn - 2(m+n)}$$

$$= x^{2mn - 2mn} \quad [\text{From (i)}]$$

$$= x^0 = 1 = \text{RHS Proved.}$$

Extra Questions

1. Simplify: a) $\sqrt[3]{(a+x)^{-8}} \times (a+x)^{\frac{2}{3}}$

Ans: $\frac{1}{(a+x)^2}$

b) $\sqrt[3]{20pq^5r^6} \times \sqrt[3]{50p^4qr^{-2}}$

Ans: $10pq^2r^2$

2. Prove that: a) $\frac{1}{1+x^{a-b}+x^{c-b}} + \frac{1}{1+x^{b-c}+x^{a-c}} + \frac{1}{1+x^{c-a}+x^{b-a}} = 1$

b) $\frac{\left(a^2 - \frac{1}{b^2}\right)^a \times \left(a - \frac{1}{b}\right)^{b-a}}{\left(b^2 - \frac{1}{a^2}\right)^b \times \left(b + \frac{1}{a}\right)^{a-b}} = \left(\frac{a}{b}\right)^{a+b}$

3. Solve: a) $2x + 2-x = 4 + 4-1$

Ans: ± 2

b) $25x - 6 \text{ a } 5x + 1 + 125 = 0$

Ans: 1, 2

4. a) $a + b + c = 0$, prove that $(1 + x^a + x^{-b})^{-1} + (1 + x^b + x^{-c})^{-1} + (1 + x^c + x^{-a})^{-1} = 1$